

MUHAMMAD SUFYAN

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Professional Summary

Electrical Engineering undergraduate at PIEAS with a 3.99/4.0 CGPA and hands-on project experience spanning multi-layer PCB design, RF hardware, embedded firmware, power electronics, and ROS 2 autonomous robotics. Skilled at turning theoretical circuit design and software development into working hardware systems using C, C++, Python, MATLAB, LTspice, and KiCad.

Education

Pakistan Institute of Engineering & Applied Sciences (PIEAS) <i>B.S. Electrical Engineering – CGPA: 3.99/4.0</i>	Sep. 2025 – Present <i>Islamabad, Pakistan</i>
Federal College of Science and Commerce Attock <i>HSSC, Pre-Engineering – Grade A</i>	Aug. 2023 – Jun. 2025 <i>Attock, Pakistan</i>
FG Public Boys High School Sanjwal Cantt <i>SSC, Science – Grade A</i>	Mar. 2021 – Jun. 2023 <i>Sanjwal Cantt Attock, Pakistan</i>

Experience

ROS 2 Software Intern ArboTrix Lab <i>Islamabad, Pakistan</i>	Jul. 2026 – Sep. 2026
– Developed autonomous warehouse mobile robot simulations and ROS 2 mission control packages, featuring multi-state delivery workflows, automated battery management, a real-time Flask web dashboard for remote telemetry and inventory administration, and testing environments in Gazebo and RViz2. – Programmed custom Python and C++ nodes for sensor integration, motor control, and live telemetry tracking.	
Volunteer Management Intern Alkhidmat Foundation Pakistan <i>Attock, Pakistan</i>	Jul. 2026 – Aug. 2026
– Coordinated community outreach surveys, youth digital skills training sessions, and emergency response reporting for regional humanitarian programs.	
C++ Intern Arch Technologies <i>Islamabad, Pakistan</i>	Feb. 2026 – Mar. 2026
– Designed modular command-line software applications in C++ applying object-oriented programming (OOP) and structured logic paradigms.	
Python Intern CodeAlpha <i>Islamabad, Pakistan</i>	Feb. 2026
– Engineered clean Python utility tools, including interactive command-line games and an object-oriented ATM management simulation.	

Projects

- **STM32WB55 Bluetooth LE 4-Layer PCB:** Designed a custom 4-layer BLE hardware board in KiCad following manufacturer layout guidelines, integrating RF matching networks, filtering, and power management planes.
- **ESP32-C3 IoT Sensor Board:** Developed a 4-layer embedded sensor board featuring onboard Li-ion battery charging circuitry, environmental sensor interfaces, and microSD card data logging.
- **DC-DC Power Converter PCBs:** Designed high-efficiency buck and boost switching regulator boards with a focus on minimizing layout parasitics, thermal optimization, and voltage stability.
- **C++ Hotel Room Management System:** Built a console application in C++ to manage room bookings, guest data tracking, availability checking, and automated billing calculations.
- **Residential House Wiring & Load Simulation:** Designed a complete residential electrical distribution network, calculated room loads, and simulated circuit behavior and backup systems using Multisim and MATLAB.
- **Arduino Embedded Systems Suite:** Implemented practical microcontroller projects including a smart door lock, an ultrasonic parking assistant, and multi-sensor monitoring nodes.
- **ESP32 Bluetooth/Wi-Fi Dual-Mode RC Car:** Built a smartphone-controlled robotic rover featuring an L298N motor driver, software PWM motor ramping, and an isolated dual-power supply setup.
- **Transistor Small-Signal Analyzer (CS50P):** Programmed a Python CLI utility for automated transistor operating-point determination and AC small-signal parameter calculations.
- **Power Supply Design & Circuit Analysis:** Designed a regulated 5.9V AC-DC power supply and verified transformer sizing and load distribution via LTspice, MATLAB, and Multisim.
- **Agentra – Multi-Model AI Agent (Hackathon):** Co-developed a LangGraph ReAct multi-agent workflow featuring LLM routing, real-time web search via Tavily API, and a custom Streamlit user interface.

Skills

PCB & Hardware	KiCad, Altium Designer, Proteus, Multi-layer PCB Layout
Embedded Systems	Arduino, ESP32, STM32, Sensor Interfacing, Power Management Systems
Simulation & Tools	MATLAB, LTspice, Multisim, ROS 2, Gazebo, Git, GitHub, LaTeX
Programming	C, C++, Python, Object-Oriented Programming (OOP)

Certifications

Harvard CS50P (Python) | Altium Designer PCB Design | Simplilearn PCB Design | Simplilearn MATLAB | MATLAB Onramp | Anthropic AI Fluency | HP LIFE AI for Business | Semrush Academy AI Visibility Toolkit Certification

Honours & Awards

- **HSSC Academic Honor (College Level):** 1st Position in Pre-Engineering cohort (1025/1100 marks), Federal College of Science and Commerce Attock.
- **SSC Academic Excellence Award (School Level):** Secured top standing with 985/1100 marks, FG Public Boys High School Sanjwal Cantt.
- **Souvenir Award for Performance:** Awarded at PIEAS Open House 2026 for event logistics and coordination support.
- **Runner-Up, Genesis 3.0 Pitch Competition:** Presented an innovative maglev propulsion concept for electric vehicles at PIEAS, Nov. 2025.

Leadership & Activities

Management Team Member: IEEE PIEAS Student Branch, InnovateX PIEAS, and PIEAS Open House 2026 Logistics Team.

Campus Ambassador: National Youth Leadership Program (NYLP), Skillistan Ventures, The Leap Pakistan, and Paragon Study Abroad Expo.

Languages & Interests

Languages: English, Urdu, Punjabi

Interests: PCB Design, RF Systems, Embedded Systems, Power Electronics, ROS 2 Robotics, Cricket, Badminton.